Rotational feedbacks in the elastic fingerprint problem: the static theory of Al-Attar et al. (2024) and a dynamic extension

Working notes (prepared with Claude) — to be checked

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1 Purpose and conventions

Section 2 restates the theory of Al-Attar et al. (2024), hereafter AA24, needed for rotational feedbacks. It covers the non-rotating loading problem and its reciprocity theorems (their §2, abridged where not needed) and the rotational theory in full (their §3), together with the spherically symmetric implementation (their §4.1). The aim is that this section is a *superset* of AA24: every equation of theirs that is used appears with its original number, and intermediate steps and remarks that they omit are added in shaded **Clarification** boxes. Clarifications are my reading of the paper and should be checked.

Section 3 then replaces the static Euler equation with the linearised Liouville (dynamic Euler) equation and repeats the derivation in the same style.

Equation tags. Tags such as (3.12) reproduce AA24's numbering. Tags such as (3.12a) are added intermediate steps. Tags such as (D3.15) are the dynamic counterparts of AA24's equations. Tags such as (P1) concern the `pyslfp` implementation.

Notation. This follows AA24 with one change: the ocean function is written $\mathcal{C}$ rather than C, to avoid a clash with moments of inertia. Principal moments are $I_1 \le I_2 < I_3$, as in AA24. The spherically symmetric implementation uses $I_1 = I_2$. ΔSL is the same quantity as ξ in Heathcote et al.

2 Static theory, following Al-Attar et al. (2024)

2.1 Equations of motion for static loading (AA24 §2.1)

A non-rotating elastic earth model occupies volume M with surface ∂M, and is initially in hydrostatic equilibrium with gravitational potential Φ. Its response to a surface load σ is governed by the weak form (Al-Attar & Tromp 2014, §2.3.2)

$$A(\mathbf{u}, \phi \,|\, \mathbf{u}', \phi') + \int_{\partial M} (\mathbf{u}' \cdot \nabla\Phi + \phi')\, \sigma \, \mathrm{d}S = 0. \tag{2.1}$$

Here $\mathbf{u}$ is the displacement, ϕ the Eulerian gravitational potential perturbation, and $(\mathbf{u}', \phi')$ test functions. $(\mathbf{u}, \phi)$ is a solution if eq. (2.1) holds for all test functions. A is the elastic–gravitational bilinear form (Al-Attar & Tromp 2014, eq. 2.52) with two properties:

$$A(\mathbf{u}, \phi \,|\, \mathbf{u}', \phi') = A(\mathbf{u}', \phi' \,|\, \mathbf{u}, \phi), \tag{2.2}$$

where displacement and potential are swapped *as pairs*, and

$$A(\mathbf{u}, \phi \,|\, \mathbf{u}', \phi') = 0 \quad \text{for all } (\mathbf{u}', \phi') \quad \text{whenever} \quad \mathbf{u} = \mathbf{a} + \mathbf{b} \times \mathbf{x}, \;\; \phi = -(\mathbf{a} + \mathbf{b} \times \mathbf{x}) \cdot \nabla\Phi, \tag{2.3–2.4}$$

for arbitrary constant $\mathbf{a}, \mathbf{b}$ (linearised rigid-body motions). Taking test functions of that form, eq. (2.5), eq. (2.1) reduces to a trivial identity. By the Fredholm alternative the problem is therefore solvable for any σ, with the solution unique up to a rigid-body motion.

> **Clarification.** The surface term is the virtual work of the load. A surface mass density σ feels the gravitational force $-\sigma\nabla\Phi$ per unit area, paired with $\mathbf{u}'$. It is also a source in Poisson's equation for ϕ, paired with ϕ'. With a rigid-body test function, $\mathbf{u}' \cdot \nabla\Phi + \phi' = \mathbf{u}' \cdot \nabla\Phi - \mathbf{u}' \cdot \nabla\Phi = 0$ pointwise, so the surface term vanishes for *every* σ. This is why no compatibility condition on σ arises. On the surface of a spherically symmetric model, $\nabla\Phi = g\,\hat{\mathbf{n}}$, so $\mathbf{u} \cdot \nabla\Phi = g\,u_r$.

2.2 A first reciprocity theorem (AA24 §2.2)

Let $(\mathbf{u}, \phi)$ solve the problem for σ, and $(\mathbf{u}^\dagger, \phi^\dagger)$ for $\sigma^\dagger$. Using each solution as the test function in the other problem gives

$$A(\mathbf{u}, \phi \,|\, \mathbf{u}^\dagger, \phi^\dagger) + \int_{\partial M} (\mathbf{u}^\dagger \cdot \nabla\Phi + \phi^\dagger)\, \sigma \, \mathrm{d}S = 0, \tag{2.6}$$

$$A(\mathbf{u}^\dagger, \phi^\dagger \,|\, \mathbf{u}, \phi) + \int_{\partial M} (\mathbf{u} \cdot \nabla\Phi + \phi)\, \sigma^\dagger \, \mathrm{d}S = 0. \tag{2.7}$$

Subtracting and using eq. (2.2),

$$\int_{\partial M} (\mathbf{u}^\dagger \cdot \nabla\Phi + \phi^\dagger)\, \sigma \, \mathrm{d}S = \int_{\partial M} (\mathbf{u} \cdot \nabla\Phi + \phi)\, \sigma^\dagger \, \mathrm{d}S, \tag{2.8}$$

which is the reciprocity theorem of Tromp & Mitrovica (1999).

2.3 Reformulation in terms of sea level (AA24 §2.3)

In hydrostatic sea-level theory,

$$\Delta\mathrm{SL} = -\frac{1}{g}(\mathbf{u}\cdot\nabla\Phi + \phi) + \frac{\Phi_g}{g}, \tag{2.9}$$

where the constant Φ_g is fixed by conservation of mass,

$$\int_{\partial M} \sigma \,\mathrm{d}S = 0, \qquad \sigma = \rho_w \mathcal{C}\,\Delta\mathrm{SL} + \zeta, \tag{2.10–2.11}$$

with direct load ζ. For ice, $\zeta = \rho_i(1-\mathcal{C})\Delta I$ (2.12). For two solutions sharing the same ocean function,

$$\mathbf{u}\cdot\nabla\Phi + \phi = -g\,\Delta\mathrm{SL} + \Phi_g, \qquad \mathbf{u}^\dagger\cdot\nabla\Phi + \phi^\dagger = -g\,\Delta\mathrm{SL}^\dagger + \Phi_g^\dagger, \tag{2.13}$$

and eq. (2.8) becomes

$$\int_{\partial M} (-g\,\Delta\mathrm{SL}^\dagger + \Phi_g^\dagger)\,\sigma \,\mathrm{d}S = \int_{\partial M} (-g\,\Delta\mathrm{SL} + \Phi_g)\,\sigma^\dagger \,\mathrm{d}S. \tag{2.14}$$

The constant terms vanish by mass conservation, giving $\int_{\partial M} \Delta\mathrm{SL}^\dagger\sigma\,\mathrm{d}S = \int_{\partial M} \Delta\mathrm{SL}\,\sigma^\dagger\,\mathrm{d}S$ (2.15). Substituting the load decompositions (2.16) and cancelling the symmetric ocean terms,

$$\int_{\partial M} \Delta\mathrm{SL}^\dagger\,\zeta\,\mathrm{d}S = \int_{\partial M} \Delta\mathrm{SL}\,\zeta^\dagger\,\mathrm{d}S. \tag{2.17}$$

> **Clarification.** (i) $\int \Phi_g^\dagger\sigma\,\mathrm{d}S = \Phi_g^\dagger\int\sigma\,\mathrm{d}S = 0$ because $\Phi_g^\dagger$ is a constant. (ii) The cancelled terms are $\rho_w\int\mathcal{C}\,\Delta\mathrm{SL}^\dagger\Delta\mathrm{SL}\,\mathrm{d}S$ on both sides. This requires the *same* $\mathcal{C}$ in both problems, i.e. fixed shorelines. (iii) The factor $-g$ is common to both sides and divides out.

Green's function and kernels (AA24 §§2.4–2.6). Writing $\Delta\mathrm{SL}(\mathbf{x}) = \int_{\partial M} G(\mathbf{x},\mathbf{x}')\zeta(\mathbf{x}')\,\mathrm{d}S_{\mathbf{x}'}$ (2.18) and taking $\zeta^\dagger = \delta(\cdot,\mathbf{x})$ in eq. (2.17) gives $G(\mathbf{x},\mathbf{x}') = G(\mathbf{x}',\mathbf{x})$ (2.21). For a functional J with

$$\delta J = \int_{\partial M} \zeta^\dagger\,\delta\Delta\mathrm{SL}\,\mathrm{d}S, \tag{2.25}$$

eq. (2.17) gives

$$\delta J = \int_{\partial M} \Delta\mathrm{SL}^\dagger\,\delta\zeta\,\mathrm{d}S. \tag{2.26}$$

Here $\Delta\mathrm{SL}^\dagger$ (the adjoint sea level) solves the ordinary fingerprint problem with direct load $\zeta^\dagger$ (the adjoint load). The ice-thickness kernel is $K = \rho_i(1-\mathcal{C})\Delta\mathrm{SL}^\dagger$ (2.28).

2.4 The generalised loading problem (AA24 §2.7)

For observables that depend directly on $\mathbf{u}$ or ϕ,

$$A(\mathbf{u},\phi\,|\,\mathbf{u}',\phi') + \int_{\partial M} (\mathbf{u}'\cdot\nabla\Phi + \phi')\,\sigma\,\mathrm{d}S + \int_{\partial M} (\mathbf{u}'\cdot\mathbf{t} + \zeta_\phi\,\phi')\,\mathrm{d}S = 0, \tag{2.29}$$

which is solvable if and only if

$$\int_{\partial M} (\mathbf{t} - \zeta_\phi\nabla\Phi)\cdot(\mathbf{a} + \mathbf{b}\times\mathbf{x})\,\mathrm{d}S = 0 \tag{2.30}$$

for all $\mathbf{a}, \mathbf{b}$. Pairing solutions for $(\sigma, \mathbf{t}, \zeta_\phi)$ and $(\sigma^\dagger, \mathbf{t}^\dagger, \zeta_\phi^\dagger)$,

$$\int_{\partial M} (\mathbf{u}^\dagger \cdot \nabla\Phi + \phi^\dagger)\sigma \,\mathrm{d}S + \int_{\partial M} (\mathbf{t} \cdot \mathbf{u}^\dagger + \zeta_\phi \phi^\dagger) \,\mathrm{d}S = \int_{\partial M} (\mathbf{u} \cdot \nabla\Phi + \phi)\sigma^\dagger \,\mathrm{d}S + \int_{\partial M} (\mathbf{t}^\dagger \cdot \mathbf{u} + \zeta_\phi^\dagger \phi) \,\mathrm{d}S, \tag{2.31}$$

$$\int_{\partial M} \Delta\mathrm{SL}^\dagger \zeta \,\mathrm{d}S - \frac{1}{g}\int_{\partial M} (\mathbf{t} \cdot \mathbf{u}^\dagger + \zeta_\phi \phi^\dagger) \,\mathrm{d}S = \int_{\partial M} \Delta\mathrm{SL}\, \zeta^\dagger \,\mathrm{d}S - \frac{1}{g}\int_{\partial M} (\mathbf{t}^\dagger \cdot \mathbf{u} + \zeta_\phi^\dagger \phi) \,\mathrm{d}S. \tag{2.32}$$

For a functional $J(\Delta\mathrm{SL}, \mathbf{u}, \phi)$, write its first-order perturbation as

$$\delta J = \int_{\partial M} \zeta^\dagger \,\delta\Delta\mathrm{SL} \,\mathrm{d}S - \frac{1}{g}\int_{\partial M} (\mathbf{t}^\dagger \cdot \delta\mathbf{u} + \zeta_\phi^\dagger \,\delta\phi) \,\mathrm{d}S \tag{2.33}$$

for suitable $(\zeta^\dagger, \mathbf{t}^\dagger, \zeta_\phi^\dagger)$. Eq. (2.32), applied with the forward problem $(\delta\Delta\mathrm{SL}, \delta\zeta, \mathbf{t} = \mathbf{0}, \zeta_\phi = 0)$, then gives

$$\delta J = \int_{\partial M} \Delta\mathrm{SL}^\dagger \,\delta\zeta \,\mathrm{d}S, \tag{2.34}$$

where $\Delta\mathrm{SL}^\dagger$ solves the generalised problem with loads $(\zeta^\dagger, \mathbf{t}^\dagger, \zeta_\phi^\dagger)$. It is well defined provided J is invariant under rigid-body motions, AA24 eq. (2.35).

2.5 Review of rotational dynamics (AA24 §3.1)

A model rotating steadily with angular velocity $\boldsymbol{\Omega}$ satisfies

$$\boldsymbol{\Omega} \times (\mathbf{I}\boldsymbol{\Omega}) = \mathbf{0}, \tag{3.1}$$

so $\boldsymbol{\Omega}$ lies along a principal axis, taken to be that of the largest moment, I_3. After deformation, $\mathbf{I} \to \mathbf{I} + \mathbf{i}$ and $\boldsymbol{\Omega} \to \boldsymbol{\Omega} + \boldsymbol{\omega}$. To first order, with $\mathbf{j} = \mathbf{i}\boldsymbol{\Omega}$,

$$\boldsymbol{\Omega} \times \mathbf{j} + I_3\, \boldsymbol{\omega} \times \boldsymbol{\Omega} + \boldsymbol{\Omega} \times (\mathbf{I}\boldsymbol{\omega}) = \mathbf{0}. \tag{3.2}$$

Clarification. Expanding $(\boldsymbol{\Omega}+\boldsymbol{\omega}) \times [(\mathbf{I}+\mathbf{i})(\boldsymbol{\Omega}+\boldsymbol{\omega})]$ to first order gives $\boldsymbol{\Omega} \times (\mathbf{i}\boldsymbol{\Omega}) + \boldsymbol{\Omega} \times (\mathbf{I}\boldsymbol{\omega}) + \boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\Omega})$, and $\mathbf{I}\boldsymbol{\Omega} = I_3\boldsymbol{\Omega}$. Taking $\boldsymbol{\Omega}\times$ of eq. (3.2) and using $\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{v}) = \boldsymbol{\Omega}(\boldsymbol{\Omega} \cdot \mathbf{v}) - \Omega^2\mathbf{v} = -\Omega^2\mathbf{v}_\perp$:

$$-\Omega^2\mathbf{j}_\perp + I_3\Omega^2\boldsymbol{\omega}_\perp - \Omega^2(\mathbf{I}\boldsymbol{\omega})_\perp = \mathbf{0}. \tag{3.2a}$$

Here $\boldsymbol{\Omega} \times (\boldsymbol{\omega} \times \boldsymbol{\Omega}) = \Omega^2\boldsymbol{\omega} - \boldsymbol{\Omega}(\boldsymbol{\Omega} \cdot \boldsymbol{\omega}) = \Omega^2\boldsymbol{\omega}_\perp$ was used. Because $\boldsymbol{\Omega}$ is along a principal axis, $\mathbf{I}$ maps the equatorial plane to itself, so $(\mathbf{I}\boldsymbol{\omega})_\perp = \mathbf{I}\boldsymbol{\omega}_\perp$. This gives eq. (3.3).

$$(I_3 - \mathbf{I})\boldsymbol{\omega}_\perp = \mathbf{j}_\perp. \tag{3.3}$$

Equation (3.2) contains no information about $\boldsymbol{\omega}_\parallel$. Conservation of angular momentum,

$$\|(\mathbf{I}+\mathbf{i})(\boldsymbol{\Omega}+\boldsymbol{\omega})\|^2 = \|\mathbf{I}\boldsymbol{\Omega}\|^2, \tag{3.4}$$

gives to first order

$$-I_3\,\boldsymbol{\omega}_\parallel = \mathbf{j}_\parallel. \tag{3.5}$$

Clarification. (i) First order in eq. (3.4): $2(\mathbf{I}\boldsymbol{\Omega}) \cdot (\mathbf{I}\boldsymbol{\omega} + \mathbf{i}\boldsymbol{\Omega}) = 0$. Since $\mathbf{I}\boldsymbol{\Omega} = I_3\boldsymbol{\Omega}$ and $\mathbf{I}$ is symmetric, $(\mathbf{I}\boldsymbol{\Omega}) \cdot (\mathbf{I}\boldsymbol{\omega}) = I_3^2\,\boldsymbol{\Omega} \cdot \boldsymbol{\omega}$ and $(\mathbf{I}\boldsymbol{\Omega}) \cdot \mathbf{j} = I_3\boldsymbol{\Omega} \cdot \mathbf{j}$. Hence $I_3\boldsymbol{\omega}_\parallel + \mathbf{j}_\parallel = \mathbf{0}$. (ii) Only the *magnitude* of the angular momentum is used because $\mathbf{L}$ is fixed in inertial space, while the Earth-fixed frame in which $\boldsymbol{\omega}$ is measured may re-orient. $\|\mathbf{L}\|$ is invariant under that re-orientation. Section 3.1 shows that eq. (3.5) is also exactly the axial component of the Liouville equation.

Combining eqs. (3.3) and (3.5),

$$\tilde{\mathbf{I}}\boldsymbol{\omega} = \mathbf{j}, \qquad \tilde{\mathbf{I}} = \begin{pmatrix} I_3 - I_1 & 0 & 0 \\ 0 & I_3 - I_2 & 0 \\ 0 & 0 & -I_3 \end{pmatrix} \tag{3.6–3.7}$$

in the principal-axis basis. $\tilde{\mathbf{I}}$ is symmetric and invertible. Because $I_3 \gg I_3 - I_1 \geq I_3 - I_2$, the x_3 component of $\boldsymbol{\omega}$ is commonly neglected; the reciprocity results do not depend on this.

Clarification. Invertibility needs $I_3 > I_2$ strictly, i.e. rotation about the axis of greatest moment. The symmetry of $\tilde{\mathbf{I}}$ is the property that makes reciprocity work in §2.6. If $\boldsymbol{\omega}_\parallel$ were retained, $\boldsymbol{\omega}_\parallel = \omega_3 \hat{\mathbf{x}}_3$ would give $\psi = -\Omega\,\omega_3\, r^2 \sin^2\theta$, which has degree-0 and degree-2, order-0 parts. The equatorial components give degree 2, order 1 only (eq. P1).

The centrifugal potential and its first-order perturbation are

$$\Psi = -\tfrac{1}{2}\|\boldsymbol{\Omega} \times \mathbf{x}\|^2, \qquad \psi = -(\boldsymbol{\Omega} \times \mathbf{x}) \cdot (\boldsymbol{\omega} \times \mathbf{x}), \tag{3.8–3.9}$$

with

$$\nabla\Psi = \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{x}), \qquad \nabla\psi = \boldsymbol{\Omega} \times (\boldsymbol{\omega} \times \mathbf{x}) + \boldsymbol{\omega} \times (\boldsymbol{\Omega} \times \mathbf{x}). \tag{3.10}$$

Clarification. $\|\boldsymbol{\Omega}\times\mathbf{x}\|^2 = \Omega^2\|\mathbf{x}\|^2 - (\boldsymbol{\Omega}\cdot\mathbf{x})^2$, so $\nabla\Psi = -\Omega^2\mathbf{x} + (\boldsymbol{\Omega}\cdot\mathbf{x})\boldsymbol{\Omega} = \boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\mathbf{x})$. The centrifugal acceleration is $-\nabla\Psi$. Replacing $\boldsymbol{\Omega}$ by $\boldsymbol{\Omega} + \boldsymbol{\omega}$ and linearising gives ψ and $\nabla\psi$.

Next, the inertia perturbation must be related to the load. For an arbitrary vector $\boldsymbol{\Omega}$,

$$\boldsymbol{\Omega} \cdot (\mathbf{I}\boldsymbol{\Omega}) = \int_M \rho\, \|\boldsymbol{\Omega} \times \mathbf{x}\|^2 \, \mathrm{d}V, \tag{3.11}$$

and under a surface load σ with displacement $\mathbf{u}$,

$$\boldsymbol{\Omega} \cdot (\mathbf{i}\boldsymbol{\Omega}) = -\int_M 2\rho\, \mathbf{u} \cdot [\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{x})] \, \mathrm{d}V + \int_{\partial M} \sigma\, \|\boldsymbol{\Omega} \times \mathbf{x}\|^2 \, \mathrm{d}S. \tag{3.12}$$

Clarification. (i) Eq. (3.11) follows from $\mathbf{I} = \int_M \rho(\|\mathbf{x}\|^2 \mathbf{1} - \mathbf{x}\mathbf{x}^T)\,\mathrm{d}V$. (ii) For eq. (3.12), work in the reference configuration: mass is conserved and a particle moves from $\mathbf{x}$ to $\mathbf{x} + \mathbf{u}$, so the solid part contributes $\int_M \rho\|\boldsymbol{\Omega} \times (\mathbf{x} + \mathbf{u})\|^2 \,\mathrm{d}V$. To first order,

$$\|\boldsymbol{\Omega} \times (\mathbf{x} + \mathbf{u})\|^2 - \|\boldsymbol{\Omega} \times \mathbf{x}\|^2 \approx 2(\boldsymbol{\Omega} \times \mathbf{x}) \cdot (\boldsymbol{\Omega} \times \mathbf{u}) = 2\mathbf{u} \cdot [(\boldsymbol{\Omega} \times \mathbf{x}) \times \boldsymbol{\Omega}] = -2\mathbf{u} \cdot [\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{x})].$$

The surface load adds mass $\sigma\,\mathrm{d}S$ at $\mathbf{x} \in \partial M$. (iii) For a fluid outer core the Lagrangian $\mathbf{u}$ is not well defined; AA24 note that an Eulerian treatment there (Dahlen 1974) leaves the results unchanged.

Replacing $\boldsymbol{\Omega}$ by $\boldsymbol{\Omega} + \boldsymbol{\omega}'$ in eq. (3.12) and keeping the part linear in $\boldsymbol{\omega}'$,

$$\boldsymbol{\omega}' \cdot \mathbf{j} = -\int_M \rho\, \mathbf{u} \cdot [\boldsymbol{\omega}' \times (\boldsymbol{\Omega} \times \mathbf{x}) + \boldsymbol{\Omega} \times (\boldsymbol{\omega}' \times \mathbf{x})] \, \mathrm{d}V + \int_{\partial M} \sigma\, (\boldsymbol{\Omega} \times \mathbf{x}) \cdot (\boldsymbol{\omega}' \times \mathbf{x}) \, \mathrm{d}S, \tag{3.13}$$

and comparing with eqs. (3.8–3.9)–(3.10),

$$\boxed{\boldsymbol{\omega}' \cdot \mathbf{j} + \int_M \rho\, \mathbf{u} \cdot \nabla\psi' \, \mathrm{d}V + \int_{\partial M} \sigma\, \psi' \, \mathrm{d}S = 0} \tag{3.14}$$

where ψ' is the centrifugal perturbation associated with $\boldsymbol{\omega}'$, while $\mathbf{j}$ is associated with $(\sigma, \mathbf{u})$.

Clarification. In eq. (3.12) the $\boldsymbol{\Omega}$ on the *left* is a dummy vector, but $\mathbf{i}$ does not depend on it. Replacing $\boldsymbol{\Omega} \to \boldsymbol{\Omega} + \boldsymbol{\omega}'$, the linear part of the left side is $\boldsymbol{\omega}' \cdot \mathbf{i}\boldsymbol{\Omega} + \boldsymbol{\Omega} \cdot \mathbf{i}\boldsymbol{\omega}' = 2\boldsymbol{\omega}' \cdot \mathbf{j}$, using the symmetry of $\mathbf{i}$. The right side gives twice the integrands shown, and the factor 2 cancels. Eq. (3.14) is purely a statement about the mass distribution at one instant. This is why it carries over unchanged to the dynamic problem in §3.

Clarification. In a spherically symmetric calculation, $\mathbf{I}$ is taken to be the *observed* inertia tensor (Mitrovica et al. 2005), even though the reference model is spherical. More generally, a rotating hydrostatic reference state would have $\nabla(\Phi+\Psi)$ in place of $\nabla\Phi$ and corresponding terms in A. As I read it, AA24 keep Ψ only to generate ψ; the symmetry arguments are the same either way.

2.6 Reciprocity theorems including rotational feedbacks (AA24 §3.2)

The static loading problem in a rotating model, in weak form, is

$$\boxed{A(\mathbf{u},\phi\,|\,\mathbf{u}',\phi') + \int_{\partial M}(\mathbf{u}'\cdot\nabla\Phi+\phi')\,\sigma\,\mathrm{d}S + \int_M \rho\,\mathbf{u}'\cdot\nabla\psi\,\mathrm{d}V + (\tilde{\mathbf{I}}\boldsymbol{\omega}-\mathbf{j})\cdot\boldsymbol{\omega}' = 0} \tag{3.15}$$

for all $(\mathbf{u}',\phi',\boldsymbol{\omega}')$. The ψ term is a tidal body force. The test function $\boldsymbol{\omega}'$ enforces eq. (3.6–3.7); $\mathbf{j}$ depends implicitly on $\mathbf{u}$ and σ.

Clarification. (i) The body force per unit volume is $-\rho\nabla\psi$. Its virtual work enters with the same sign convention as the load term in eq. (2.1). (ii) The unknowns are now $(\mathbf{u},\phi,\boldsymbol{\omega})$, and $\boldsymbol{\omega}'$ acts as a Lagrange multiplier for the rotational constraint. (iii) The problem is coupled: ψ depends on $\boldsymbol{\omega}$, $\mathbf{j}$ depends on $\mathbf{u}$ and σ, and σ depends on ΔSL, which depends on ψ (eq. 3.18). This is why it is solved iteratively.

Let $(\mathbf{u}^\dagger,\phi^\dagger,\boldsymbol{\omega}^\dagger)$ solve the problem for $\sigma^\dagger$. Then

$$\int_{\partial M}(\mathbf{u}^\dagger\cdot\nabla\Phi+\phi^\dagger)\sigma\,\mathrm{d}S+\int_M\rho\,\mathbf{u}^\dagger\cdot\nabla\psi\,\mathrm{d}V+\mathbf{j}^\dagger\cdot\boldsymbol{\omega}=\int_{\partial M}(\mathbf{u}\cdot\nabla\Phi+\phi)\sigma^\dagger\,\mathrm{d}S+\int_M\rho\,\mathbf{u}\cdot\nabla\psi^\dagger\,\mathrm{d}V+\mathbf{j}\cdot\boldsymbol{\omega}^\dagger. \tag{3.16}$$

Clarification. Write eq. (3.15) for σ with test functions $(\mathbf{u}^\dagger,\phi^\dagger,\boldsymbol{\omega}^\dagger)$, and for $\sigma^\dagger$ with test functions $(\mathbf{u},\phi,\boldsymbol{\omega})$, then subtract. The A terms cancel by eq. (2.2), and $\boldsymbol{\omega}^\dagger\cdot\tilde{\mathbf{I}}\boldsymbol{\omega}=\boldsymbol{\omega}\cdot\tilde{\mathbf{I}}\boldsymbol{\omega}^\dagger$ cancels by symmetry of $\tilde{\mathbf{I}}$. What remains is $[\ldots]-\mathbf{j}\cdot\boldsymbol{\omega}^\dagger=[\ldots]^\dagger-\mathbf{j}^\dagger\cdot\boldsymbol{\omega}$. Moving the $\mathbf{j}$ terms across gives eq. (3.16).

Using eq. (3.14) twice,

$$\boxed{\int_{\partial M}(\mathbf{u}^\dagger\cdot\nabla\Phi+\phi^\dagger+\psi^\dagger)\,\sigma\,\mathrm{d}S=\int_{\partial M}(\mathbf{u}\cdot\nabla\Phi+\phi+\psi)\,\sigma^\dagger\,\mathrm{d}S} \tag{3.17}$$

Clarification. Apply eq. (3.14) with $(\mathbf{j},\mathbf{u},\sigma)$ from the † problem and $\psi'=\psi$: $\mathbf{j}^\dagger\cdot\boldsymbol{\omega}=-\int\rho\,\mathbf{u}^\dagger\cdot\nabla\psi\,\mathrm{d}V-\int\sigma^\dagger\psi\,\mathrm{d}S$. Similarly $\mathbf{j}\cdot\boldsymbol{\omega}^\dagger=-\int\rho\,\mathbf{u}\cdot\nabla\psi^\dagger\,\mathrm{d}V-\int\sigma\psi^\dagger\,\mathrm{d}S$. Substituting into eq. (3.16), both volume integrals cancel on each side, and the surface terms rearrange into eq. (3.17).

With rotation, the sea surface is an equipotential of $\Phi+\Psi$, so

$$\Delta\mathrm{SL}=-\frac{1}{g}(\mathbf{u}\cdot\nabla\Phi+\phi+\psi)+\frac{\Phi_g}{g}, \tag{3.18}$$

and the argument of §2.3 gives

$$\boxed{\int_{\partial M}\Delta\mathrm{SL}^\dagger\,\zeta\,\mathrm{d}S=\int_{\partial M}\Delta\mathrm{SL}\,\zeta^\dagger\,\mathrm{d}S} \tag{3.19}$$

in the same form as eq. (2.17). Symmetry of the Green's function and the kernel results of §2.3 therefore carry over unchanged.

2.7 Generalised problem with rotation (AA24 §3.2, continued)

$$A(\mathbf{u},\phi\,|\,\mathbf{u}',\phi')+\int_{\partial M}(\mathbf{u}'\cdot\nabla\Phi+\phi')\sigma\,\mathrm{d}S+\int_{\partial M}(\mathbf{t}\cdot\mathbf{u}'+\zeta_\phi\phi')\,\mathrm{d}S+\int_M\rho\,\mathbf{u}'\cdot\nabla\psi\,\mathrm{d}V+(\tilde{\mathbf{I}}\boldsymbol{\omega}-\mathbf{j}+\mathbf{k})\cdot\boldsymbol{\omega}'=0, \tag{3.20}$$

where $\mathbf{k}$ is an additional forcing for observables that depend directly on $\boldsymbol{\omega}$. The direct gravitational load ζ_ϕ also contributes to the inertia perturbation, so eq. (3.14) generalises to

$$\boldsymbol{\omega}'\cdot\mathbf{j}+\int_M\rho\,\mathbf{u}\cdot\nabla\psi'\,\mathrm{d}V+\int_{\partial M}(\sigma+\zeta_\phi)\,\psi'\,\mathrm{d}S=0. \tag{3.21}$$

Clarification. As I read it, ζ_ϕ acts as a surface mass in Poisson's equation (it pairs with ϕ' exactly as σ does) but exerts no force on $\mathbf{u}$. The inertia perturbation inferred from the perturbed potential therefore includes $\int\zeta_\phi\|\boldsymbol{\Omega}\times\mathbf{x}\|^2\,\mathrm{d}S$. This is what the spectral implementation does automatically, since $\boldsymbol{\omega}_\perp$ is computed from ϕ_{21} (eq. P2).

The same argument as before then gives

$$\begin{aligned}\int_{\partial M}\Delta\mathrm{SL}^\dagger\zeta\,\mathrm{d}S-\frac{1}{g}\int_{\partial M}[\mathbf{t}\cdot\mathbf{u}^\dagger+\zeta_\phi(\phi^\dagger+\psi^\dagger)]\,\mathrm{d}S-\frac{1}{g}\,\mathbf{k}\cdot\boldsymbol{\omega}^\dagger\\ =\int_{\partial M}\Delta\mathrm{SL}\,\zeta^\dagger\,\mathrm{d}S-\frac{1}{g}\int_{\partial M}[\mathbf{t}^\dagger\cdot\mathbf{u}+\zeta_\phi^\dagger(\phi+\psi)]\,\mathrm{d}S-\frac{1}{g}\,\mathbf{k}^\dagger\cdot\boldsymbol{\omega}.\end{aligned} \tag{3.22}$$

Clarification. Subtracting the two instances of eq. (3.20) gives the analogue of eq. (3.16), with extra terms $\int(\mathbf{t}\cdot\mathbf{u}^\dagger+\zeta_\phi\phi^\dagger)+\mathbf{k}\cdot\boldsymbol{\omega}^\dagger$ on the left and their †-counterparts on the right. Using eq. (3.21) in place of eq. (3.14) produces $\int\zeta_\phi\psi^\dagger$ and $\int\zeta_\phi^\dagger\psi$. Then substitute eq. (3.18), use $\int\sigma\,\mathrm{d}S=0$, divide by $-g$, and cancel the symmetric ocean terms.

Accordingly, for a functional $J(\Delta\mathrm{SL},\mathbf{u},\phi,\boldsymbol{\omega})$ write

$$\delta J=\int_{\partial M}\zeta^\dagger\,\delta\Delta\mathrm{SL}\,\mathrm{d}S-\frac{1}{g}\int_{\partial M}[\mathbf{t}^\dagger\cdot\delta\mathbf{u}+\zeta_\phi^\dagger(\delta\phi+\delta\psi)]\,\mathrm{d}S-\frac{1}{g}\,\mathbf{k}^\dagger\cdot\delta\boldsymbol{\omega}; \tag{3.22a}$$

eq. (3.22) then gives $\delta J=\int_{\partial M}\Delta\mathrm{SL}^\dagger\,\delta\zeta\,\mathrm{d}S$, with $\Delta\mathrm{SL}^\dagger$ from the generalised rotating problem with loads $(\zeta^\dagger,\mathbf{t}^\dagger,\zeta_\phi^\dagger,\mathbf{k}^\dagger)$.

Since $\psi=-[\mathbf{x}\times(\boldsymbol{\Omega}\times\mathbf{x})]\cdot\boldsymbol{\omega}$ (scalar triple product), the $\zeta_\phi\psi$ terms can be absorbed into the $\mathbf{k}$ terms:

$$\begin{aligned}\int_{\partial M}\Delta\mathrm{SL}^\dagger\zeta\,\mathrm{d}S-\frac{1}{g}\int_{\partial M}(\mathbf{t}\cdot\mathbf{u}^\dagger+\zeta_\phi\phi^\dagger)\,\mathrm{d}S-\frac{1}{g}\Big(\mathbf{k}-\int_{\partial M}\zeta_\phi\,\mathbf{x}\times(\boldsymbol{\Omega}\times\mathbf{x})\,\mathrm{d}S\Big)\cdot\boldsymbol{\omega}^\dagger\\ =\int_{\partial M}\Delta\mathrm{SL}\,\zeta^\dagger\,\mathrm{d}S-\frac{1}{g}\int_{\partial M}(\mathbf{t}^\dagger\cdot\mathbf{u}+\zeta_\phi^\dagger\phi)\,\mathrm{d}S-\frac{1}{g}\Big(\mathbf{k}^\dagger-\int_{\partial M}\zeta_\phi^\dagger\,\mathbf{x}\times(\boldsymbol{\Omega}\times\mathbf{x})\,\mathrm{d}S\Big)\cdot\boldsymbol{\omega}.\end{aligned} \tag{3.23}$$

2.8 Spherically symmetric implementation (AA24 §4.1, plus `pyslfp`)

With $\sigma=\sum_{lm}\sigma_{lm}Y_{lm}$ (4.1), using orthonormal real harmonics without the Condon–Shortley phase,

$$u_{lm}=h_l\sigma_{lm},\qquad\phi_{lm}=k_l\sigma_{lm}, \tag{4.2}$$

where u is the vertical displacement. AA24's k_l includes the potential of the load itself, which is non-standard. For functionals depending on $\mathbf{u}$ only through its vertical component, $\mathbf{t}=\zeta_u\nabla\Phi$ (4.3), and

$$u_{lm}=h_l\sigma_{lm}+h_l^u\zeta_{lm}^u+h_l^\phi\zeta_{lm}^\phi,\qquad\phi_{lm}=k_l\sigma_{lm}+k_l^u\zeta_{lm}^u+k_l^\phi\zeta_{lm}^\phi, \tag{4.4}$$

with $h_l=h_l^u+h_l^\phi$ and $k_l=k_l^u+k_l^\phi$ (4.5), and $g\,h_l^\phi=k_l^u$ (4.6). Tidal Love numbers h_l^T,k_l^T give the response to the body force $-\rho\nabla\psi$: $u_{lm}=h_l^T\psi_{lm}$, $\phi_{lm}=k_l^T\psi_{lm}$, where ϕ excludes ψ itself.

Clarification. (4.5): comparing eq. (2.1) with eq. (2.29) and $\mathbf{t} = \zeta_u \nabla\Phi$, a true load σ is equivalent to $\zeta_u = \zeta_\phi = \sigma$. (4.6): in eq. (2.31) with $\sigma = \sigma^\dagger = 0$, take a unit ζ_u harmonic in one problem and a unit ζ_ϕ harmonic in the other. Then $\int \mathbf{t}\cdot\mathbf{u}^\dagger = g\, h_l^\phi$ and $\int \zeta_\phi^\dagger \phi = k_l^u$.

Rotational terms (not written out in AA24). Write $\boldsymbol{\phi}_{21} = (\phi_{21}^c, \phi_{21}^s)^T$ for the $\cos\varphi$ and $\sin\varphi$ coefficients, with fields evaluated at radius b. With $\boldsymbol{\Omega} = \Omega\hat{\mathbf{x}}_3$ and $\boldsymbol{\omega}_\perp = (\omega_1, \omega_2)$, eq. (3.8–3.9) gives $\psi = \Omega x_3(x_1\omega_1 + x_2\omega_2) = \Omega r^2 \sin\theta\cos\theta\,(\omega_1\cos\varphi + \omega_2\sin\varphi)$. Since $Y_{21}^c = \sqrt{15/4\pi}\sin\theta\cos\theta\cos\varphi$,

$$\boldsymbol{\psi}_{21} = r\,\boldsymbol{\omega}_\perp, \qquad r = \sqrt{\tfrac{4\pi}{15}}\,\Omega b^2. \tag{P1}$$

From MacCullagh's formula, the exterior potential of a mass distribution has $\phi_{21}^c = \sqrt{12\pi/5}\,G\,\mathrm{i}_{13}/b^3$, and similarly ϕ_{21}^s with i_{23}. With $\mathbf{j}_\perp = \Omega(\mathrm{i}_{13}, \mathrm{i}_{23})$ and $I_1 = I_2$,

$$\frac{\mathbf{j}_\perp}{I_3 - I_1} = \beta\,\boldsymbol{\phi}_{21}, \qquad \beta = \sqrt{\tfrac{5}{12\pi}}\,\frac{\Omega b^3}{G(I_3 - I_1)}, \tag{P2}$$

where $\boldsymbol{\phi}_{21} = k_2\boldsymbol{\sigma}_{21} + k_2^T\boldsymbol{\psi}_{21}$. The static relation (3.20), restricted to equatorial components, is

$$\boldsymbol{\omega}_\perp = \mathbf{q}, \qquad \mathbf{q} := \beta\boldsymbol{\phi}_{21} - \frac{\mathbf{k}_\perp}{I_3 - I_1}, \tag{P3}$$

with ω_3 neglected. This is exactly the `pyslfp` update: r and β are `rotation_factor` and `inertia_factor`, and $\mathbf{k}$ is `angular_momentum_change`. The feedback strength is

$$\kappa := \beta r k_2^T = \frac{k_2^T}{k_s}, \qquad k_s = \frac{3G(I_3 - I_1)}{\Omega^2 b^5}, \tag{P4}$$

with $k_s = 0.951$ for the `pyslfp` constants and $k_2^T = 0.2986$ (PREM), so $\kappa = 0.314$.

Clarification. *MacCullagh step.* With orthonormal harmonics, the exterior potential is

$$\phi(r) = -G\sum_m \frac{4\pi}{5}\, r^{-3}\, Y_{2m}(\hat{\mathbf{x}}) \int_M \rho\, r'^2\, Y_{2m}(\hat{\mathbf{x}}')\,\mathrm{d}V',$$

and $\int \rho\, r'^2 Y_{21}^c\,\mathrm{d}V = \sqrt{15/4\pi}\int \rho\, x_1 x_3\,\mathrm{d}V = -\sqrt{15/4\pi}\,\mathrm{i}_{13}$. Hence

$$\phi_{21}^c = \frac{G}{b^3}\,\frac{4\pi}{5}\sqrt{\frac{15}{4\pi}}\,\mathrm{i}_{13} = \sqrt{\frac{12\pi}{5}}\,\frac{G\,\mathrm{i}_{13}}{b^3}.$$

Because ϕ is the potential of *all* perturbed mass (internal deformation, σ, and ζ_ϕ in the generalised problem), eq. (P2) automatically contains both terms of eq. (3.12) and the ζ_ϕ term of eq. (3.21). *Iteration.* `pyslfp` alternates: compute $\boldsymbol{\psi}_{21}$ from the current $\boldsymbol{\omega}$; compute u and ϕ from σ and ψ; update $\boldsymbol{\omega}_\perp \leftarrow \mathbf{q}$; form ΔSL from eq. (3.18); update σ.

3 Dynamic theory

This section mirrors §§2.5–2.8, with eq. (3.1) replaced by the linearised Liouville equation. The elastic response and the ocean remain instantaneous (quasi-static). The only new ingredient is rotational inertia.

3.1 Rotational dynamics with time dependence

In the Earth-fixed frame, with no external torque, the angular momentum $\mathbf{L} = (\mathbf{I} + \mathbf{i})(\boldsymbol{\Omega} + \boldsymbol{\omega})$ satisfies

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big[(\mathbf{I} + \mathbf{i})(\boldsymbol{\Omega} + \boldsymbol{\omega})\Big] + (\boldsymbol{\Omega} + \boldsymbol{\omega}) \times \Big[(\mathbf{I} + \mathbf{i})(\boldsymbol{\Omega} + \boldsymbol{\omega})\Big] = \mathbf{0}, \tag{D3.1}$$

of which eq. (3.1) is the steady, unperturbed case. To first order, with $\mathbf{j} = \mathbf{i}\boldsymbol{\Omega}$ and $\boldsymbol{\Omega}$ constant,

$$\boxed{\mathbf{I}\dot{\boldsymbol{\omega}} + \dot{\mathbf{j}} + \boldsymbol{\Omega} \times \mathbf{j} + I_3\, \boldsymbol{\omega} \times \boldsymbol{\Omega} + \boldsymbol{\Omega} \times (\mathbf{I}\boldsymbol{\omega}) = \mathbf{0}} \tag{D3.2}$$

which is eq. (3.2) plus the two time-derivative terms.

> **Clarification.** (i) The time derivative is taken in the Earth-fixed frame, which is why the transport term $(\boldsymbol{\Omega} + \boldsymbol{\omega}) \times \mathbf{L}$ appears. (ii) Torque-free: surface mass redistribution is internal to the Earth–ocean–ice system, and lunisolar torques act at nearly diurnal frequencies in this frame, outside the band considered. (iii) The relative angular momentum of the deformation, $\int \rho\, \mathbf{x} \times \dot{\mathbf{u}}\, \mathrm{d}V$, is omitted; it vanishes in a Tisserand frame and is in any case $O(\epsilon)$ (see below). Relative angular momentum of ocean currents or the atmosphere would enter as $-\mathbf{k}$ in the generalised problem.

Equatorial components. Taking $\boldsymbol{\Omega}\times$ of eq. (D3.2) and reusing the steps of eq. (3.2a),

$$\boldsymbol{\Omega} \times (\mathbf{I}\dot{\boldsymbol{\omega}}) + \boldsymbol{\Omega} \times \dot{\mathbf{j}} + \Omega^2 \Big[(I_3 - \mathbf{I})\boldsymbol{\omega}_\perp - \mathbf{j}_\perp \Big] = \mathbf{0}. \tag{D3.3a}$$

Define the *gyroscopic operator*

$$\boldsymbol{\Gamma}\mathbf{v} := \frac{1}{\Omega^2}\, \boldsymbol{\Omega} \times (\mathbf{I}\mathbf{v}), \qquad \boldsymbol{\Gamma} = \frac{1}{\Omega} \begin{pmatrix} 0 & -I_2 & 0 \\ I_1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \tag{D3.3b}$$

in the principal-axis basis. $\boldsymbol{\Gamma}$ maps into the equatorial plane and annihilates axial vectors. Then eq. (D3.3a) is

$$(I_3 - \mathbf{I})\boldsymbol{\omega}_\perp + \boldsymbol{\Gamma}\dot{\boldsymbol{\omega}} + \frac{1}{\Omega^2}\boldsymbol{\Omega} \times \dot{\mathbf{j}} = \mathbf{j}_\perp. \tag{D3.3}$$

Ordering. Let T be the timescale of the loading or response, and define $\epsilon = 1/(\Omega T)$ and $\delta = (I_3 - I_1)/I_1 \approx 3.3 \times 10^{-3}$. In eq. (D3.3), relative to $(I_3 - \mathbf{I})\boldsymbol{\omega}_\perp \sim \mathbf{j}_\perp$:

- $\boldsymbol{\Gamma}\dot{\boldsymbol{\omega}} \sim (I_1/\Omega T)\, \omega$, i.e. $O(\epsilon/\delta)$, which is $O(1)$ once $T \lesssim$ years: **retain**.
- $\Omega^{-2}\boldsymbol{\Omega} \times \dot{\mathbf{j}}$ is $O(\epsilon)$: drop.

The retained term is the only one amplified by the small dynamical ellipticity. The two regimes are separated by $1/\sigma_e$, where $\sigma_e = \Omega\delta$, so $1/\sigma_e \approx 48\,\mathrm{d}$ and $2\pi/\sigma_e = 304.5\,\mathrm{d}$ is the Euler period.

Axial component. Dotting eq. (D3.2) with $\boldsymbol{\Omega}$ removes every cross product. Using $\boldsymbol{\Omega} \cdot \mathbf{I}\dot{\boldsymbol{\omega}} = (\mathbf{I}\boldsymbol{\Omega}) \cdot \dot{\boldsymbol{\omega}} = I_3\, \boldsymbol{\Omega} \cdot \dot{\boldsymbol{\omega}}$,

$$I_3\, \dot{\boldsymbol{\omega}}_\parallel + \dot{\mathbf{j}}_\parallel = \mathbf{0} \quad \Longrightarrow \quad -I_3\, \boldsymbol{\omega}_\parallel = \mathbf{j}_\parallel, \tag{D3.5}$$

given an unperturbed initial state. This is eq. (3.5) exactly: the angular-momentum argument of the static theory is simply the axial Liouville equation, and involves no approximation.

> **Clarification.** In eq. (D3.5) the $\dot{\mathbf{j}}_\parallel$ term is *not* dropped: it is the only term balancing $I_3\dot{\boldsymbol{\omega}}_\parallel$, and the ordering argument applies to the equatorial balance only.

Combining eqs. (D3.3) and (D3.5), and noting that $\boldsymbol{\Gamma}$ has no axial part:

$$\boxed{\tilde{\mathbf{I}}\boldsymbol{\omega} + \boldsymbol{\Gamma}\dot{\boldsymbol{\omega}} = \mathbf{j}} \tag{D3.6}$$

This is the dynamic counterpart of eq. (3.6–3.7), which is recovered when $\dot{\boldsymbol{\omega}} = \mathbf{0}$. The key structural property is

$$\boldsymbol{\Gamma}^T = -\boldsymbol{\Gamma} \quad \Longleftrightarrow \quad I_1 = I_2, \tag{D3.7}$$

since $\boldsymbol{\Gamma} = \Omega^{-2}[\boldsymbol{\Omega}]_\times \mathbf{I}$ and $\boldsymbol{\Gamma}^T = -\Omega^{-2}\mathbf{I}[\boldsymbol{\Omega}]_\times$. From here on, $I_1 = I_2$.

Clarification. *In what sense $\boldsymbol{\Omega}+\boldsymbol{\omega}$ "changes direction" dynamically.* In the static theory $\boldsymbol{\omega}_\perp$ is slaved to $\mathbf{j}_\perp$: the rotation axis sits instantaneously on the equilibrium axis. In eq. (D3.6) it has inertia. When the figure axis moves, the rotation axis wobbles about it at the free (Chandler) frequency (§3.4).

Centrifugal potential and inertia identity. Eqs. (3.8–3.9)–(3.10) hold at each instant with $\boldsymbol{\omega} = \boldsymbol{\omega}(t)$. The Euler force $-\rho\dot{\boldsymbol{\omega}} \times \mathbf{x}$ in the momentum equation is non-conservative, but it is $O(\epsilon)$ relative to $\rho\nabla\psi$ and is dropped. Coriolis and inertial terms in the solid Earth, and the non-equilibrium ocean response, are likewise $O(\epsilon)$. Eqs. (3.11)–(3.14) involve only the mass distribution at one instant, so

$$\boldsymbol{\omega}' \cdot \mathbf{j}(t) + \int_M \rho\,\mathbf{u}(t) \cdot \nabla\psi'\,\mathrm{d}V + \int_{\partial M} \sigma(t)\,\psi'\,\mathrm{d}S = 0 \qquad \text{for each } t. \tag{D3.14}$$

3.2 Reciprocity with dynamic rotation

On $t \in [0, T]$, the equations of motion in weak form are

$$\boxed{A(\mathbf{u}, \phi \,|\, \mathbf{u}', \phi') + \int_{\partial M} (\mathbf{u}' \cdot \nabla\Phi + \phi')\,\sigma\,\mathrm{d}S + \int_M \rho\,\mathbf{u}' \cdot \nabla\psi\,\mathrm{d}V + (\tilde{\mathbf{I}}\boldsymbol{\omega} + \boldsymbol{\Gamma}\dot{\boldsymbol{\omega}} - \mathbf{j}) \cdot \boldsymbol{\omega}' = 0, \qquad \boldsymbol{\omega}(0) = \mathbf{0}} \tag{D3.15}$$

for all t and all test functions. At each instant, $(\mathbf{u}, \phi)$ are determined by $\sigma(t)$ and $\psi(t)$; the only memory is in $\boldsymbol{\omega}$.

Clarification. A non-zero $\boldsymbol{\omega}(0)$, representing a pre-existing free wobble, is treated at the end of this subsection.

Let $(\mathbf{u}^\dagger, \phi^\dagger, \boldsymbol{\omega}^\dagger)$ solve eq. (D3.15) for a load $\sigma^\dagger(t)$, but with the *terminal* condition $\boldsymbol{\omega}^\dagger(T) = \mathbf{0}$ instead of an initial one. Then for each t,

$$\begin{aligned}\int_{\partial M} (\mathbf{u}^\dagger \cdot \nabla\Phi + \phi^\dagger)\sigma\,\mathrm{d}S + \int_M \rho\,\mathbf{u}^\dagger \cdot \nabla\psi\,\mathrm{d}V + \mathbf{j}^\dagger \cdot \boldsymbol{\omega} + \boldsymbol{\omega}^\dagger \cdot \boldsymbol{\Gamma}\dot{\boldsymbol{\omega}} \\ = \int_{\partial M} (\mathbf{u} \cdot \nabla\Phi + \phi)\sigma^\dagger\,\mathrm{d}S + \int_M \rho\,\mathbf{u} \cdot \nabla\psi^\dagger\,\mathrm{d}V + \mathbf{j} \cdot \boldsymbol{\omega}^\dagger + \boldsymbol{\omega} \cdot \boldsymbol{\Gamma}\dot{\boldsymbol{\omega}}^\dagger.\end{aligned} \tag{D3.16}$$

Clarification. This is derived exactly as eq. (3.16). The only new terms are $\boldsymbol{\omega}^\dagger \cdot \boldsymbol{\Gamma}\dot{\boldsymbol{\omega}}$, from the σ-equation tested with $\boldsymbol{\omega}^\dagger$, and $\boldsymbol{\omega} \cdot \boldsymbol{\Gamma}\dot{\boldsymbol{\omega}}^\dagger$. They do not cancel, because $\boldsymbol{\Gamma}$ multiplies a time derivative.

Using eq. (D3.14) twice, exactly as for eq. (3.17),

$$\int_{\partial M} (\mathbf{u}^\dagger \cdot \nabla\Phi + \phi^\dagger + \psi^\dagger)\sigma\,\mathrm{d}S - \int_{\partial M} (\mathbf{u} \cdot \nabla\Phi + \phi + \psi)\sigma^\dagger\,\mathrm{d}S = \boldsymbol{\omega} \cdot \boldsymbol{\Gamma}\dot{\boldsymbol{\omega}}^\dagger - \boldsymbol{\omega}^\dagger \cdot \boldsymbol{\Gamma}\dot{\boldsymbol{\omega}} = -\frac{\mathrm{d}}{\mathrm{d}t}(\boldsymbol{\omega}^\dagger \cdot \boldsymbol{\Gamma}\boldsymbol{\omega}). \tag{D3.17a}$$

Clarification. By eq. (D3.7), $\boldsymbol{\omega} \cdot \boldsymbol{\Gamma}\dot{\boldsymbol{\omega}}^\dagger = (\boldsymbol{\Gamma}^T\boldsymbol{\omega}) \cdot \dot{\boldsymbol{\omega}}^\dagger = -\dot{\boldsymbol{\omega}}^\dagger \cdot \boldsymbol{\Gamma}\boldsymbol{\omega}$. Hence the right side is $-(\dot{\boldsymbol{\omega}}^\dagger \cdot \boldsymbol{\Gamma}\boldsymbol{\omega} + \boldsymbol{\omega}^\dagger \cdot \boldsymbol{\Gamma}\dot{\boldsymbol{\omega}})$, a total derivative. For $I_1 \neq I_2$ the symmetric part of $\boldsymbol{\Gamma}$ leaves a non-derivative remainder, and this argument fails.

Integrating over $[0, T]$, the boundary term $[\boldsymbol{\omega}^\dagger \cdot \boldsymbol{\Gamma}\boldsymbol{\omega}]_0^T$ vanishes because $\boldsymbol{\omega}(0) = \mathbf{0}$ and $\boldsymbol{\omega}^\dagger(T) = \mathbf{0}$:

$$\boxed{\int_0^T \int_{\partial M} (\mathbf{u}^\dagger \cdot \nabla\Phi + \phi^\dagger + \psi^\dagger)\,\sigma\,\mathrm{d}S\,\mathrm{d}t = \int_0^T \int_{\partial M} (\mathbf{u} \cdot \nabla\Phi + \phi + \psi)\,\sigma^\dagger\,\mathrm{d}S\,\mathrm{d}t} \tag{D3.17}$$

Eq. (3.18) holds at each instant, and the steps of §2.3 are instantaneous (the ocean terms cancel pointwise in t), so

$$\boxed{\int_0^T \int_{\partial M} \Delta\mathrm{SL}^\dagger\,\zeta\,\mathrm{d}S\,\mathrm{d}t = \int_0^T \int_{\partial M} \Delta\mathrm{SL}\,\zeta^\dagger\,\mathrm{d}S\,\mathrm{d}t} \tag{D3.19}$$

The adjoint problem. The † fields satisfy the *same* equations as the forward problem, but with a terminal condition, so they are integrated backwards from T. Let $\tau = T - t$ and $\tilde{\boldsymbol{\omega}}(\tau) = \boldsymbol{\omega}^\dagger(T-\tau)$. Then $\mathrm{d}\tilde{\boldsymbol{\omega}}/\,\mathrm{d}\tau = -\dot{\boldsymbol{\omega}}^\dagger$, and eq. (D3.6) becomes

$$\tilde{\mathbf{I}}\tilde{\boldsymbol{\omega}} - \boldsymbol{\Gamma}\,\frac{\mathrm{d}\tilde{\boldsymbol{\omega}}}{\mathrm{d}\tau} = \tilde{\mathbf{j}}, \qquad \tilde{\boldsymbol{\omega}}(0) = \mathbf{0}: \tag{D3.19a}$$

a causal forward problem in τ, with $\boldsymbol{\Gamma} \to -\boldsymbol{\Gamma}$ and the time-reversed adjoint load $\tilde{\zeta}(\tau) = \zeta^\dagger(T-\tau)$.

> **Clarification.** $\boldsymbol{\Gamma} \to -\boldsymbol{\Gamma}$ is equivalent to reversing the sense of the reference rotation. Under $\boldsymbol{\Omega} \to -\boldsymbol{\Omega}$ together with $\boldsymbol{\omega} \to -\boldsymbol{\omega}$: ψ is unchanged, $\tilde{\mathbf{I}}\boldsymbol{\omega} \to -\tilde{\mathbf{I}}\boldsymbol{\omega}$, $\mathbf{j} = \mathbf{i}\boldsymbol{\Omega} \to -\mathbf{j}$, and $\boldsymbol{\Gamma}\dot{\boldsymbol{\omega}} \to +\boldsymbol{\Gamma}\dot{\boldsymbol{\omega}}$ (both $\boldsymbol{\Gamma}$ and $\dot{\boldsymbol{\omega}}$ flip). Dividing by -1 gives eq. (D3.19a), while $\Delta\mathrm{SL}$ is unchanged. This is the familiar reciprocity of rotating systems: source and receiver are exchanged and the rotation is reversed.

Green's function. By linearity and time invariance, $\Delta\mathrm{SL}(\mathbf{x},t) = \int_0^t \int_{\partial M} G_{\boldsymbol{\Gamma}}(\mathbf{x},\mathbf{x}';t-t')\,\zeta(\mathbf{x}',t')\,\mathrm{d}S_{\mathbf{x}'}\,\mathrm{d}t'$ (D2.18). Take $\zeta = \delta(\cdot,\mathbf{x}_1)\delta(t-t_1)$ and $\zeta^\dagger = \delta(\cdot,\mathbf{x}_2)\delta(t-t_2)$. By eq. (D3.19a), $\Delta\mathrm{SL}^\dagger(\mathbf{x},t) = G_{-\boldsymbol{\Gamma}}(\mathbf{x},\mathbf{x}_2;t_2-t)$, and eq. (D3.19) gives

$$G_{\boldsymbol{\Gamma}}(\mathbf{x}_2,\mathbf{x}_1;s) = G_{-\boldsymbol{\Gamma}}(\mathbf{x}_1,\mathbf{x}_2;s). \tag{D2.21}$$

Integrating over s recovers the static symmetry (2.21), because the static Green's function does not depend on $\boldsymbol{\Gamma}$.

Sensitivity kernels. If $\delta J = \int_0^T \int_{\partial M} \zeta^\dagger\,\delta\Delta\mathrm{SL}\,\mathrm{d}S\,\mathrm{d}t$, then

$$\delta J = \int_0^T \int_{\partial M} \Delta\mathrm{SL}^\dagger\,\delta\zeta\,\mathrm{d}S\,\mathrm{d}t, \qquad K(\mathbf{x},t) = \rho_i(1-\mathcal{C})\,\Delta\mathrm{SL}^\dagger(\mathbf{x},t). \tag{D2.26}$$

For an observation at a single time t_o, $\zeta^\dagger \propto \delta(t-t_o)$. The backward solution then vanishes for $t > t_o$, so the kernel correctly depends only on the loading history before t_o.

Non-zero initial state. If $\boldsymbol{\omega}(0) = \boldsymbol{\omega}_0$ (with $\boldsymbol{\omega}^\dagger(T) = \mathbf{0}$ still), the boundary term survives, and eq. (D3.19) becomes

$$\int_0^T \int_{\partial M} \Delta\mathrm{SL}\,\zeta^\dagger\,\mathrm{d}S\,\mathrm{d}t = \int_0^T \int_{\partial M} \Delta\mathrm{SL}^\dagger\,\zeta\,\mathrm{d}S\,\mathrm{d}t + \frac{1}{g}\,\boldsymbol{\omega}^\dagger(0)\cdot\boldsymbol{\Gamma}\boldsymbol{\omega}_0, \tag{D3.19b}$$

so $\partial J/\partial\boldsymbol{\omega}_0 = g^{-1}\boldsymbol{\Gamma}^T\boldsymbol{\omega}^\dagger(0)$. This gives the sensitivity to an initial free wobble, e.g. if it is treated as a nuisance parameter.

> **Clarification.** From eq. (D3.17a), the space–time difference equals $-[\boldsymbol{\omega}^\dagger\cdot\boldsymbol{\Gamma}\boldsymbol{\omega}]_0^T = \boldsymbol{\omega}^\dagger(0)\cdot\boldsymbol{\Gamma}\boldsymbol{\omega}_0$. Using eq. (3.18) and mass conservation, the left side equals $-g\big[\iint \Delta\mathrm{SL}^\dagger\zeta - \iint \Delta\mathrm{SL}\,\zeta^\dagger\big]$. Divide by $-g$ and rearrange.

3.3 Generalised dynamic problem

The generalised problem is

$$A(\mathbf{u},\phi\,|\,\mathbf{u}',\phi') + \int_{\partial M}(\mathbf{u}'\cdot\nabla\Phi+\phi')\sigma\,\mathrm{d}S + \int_{\partial M}(\mathbf{t}\cdot\mathbf{u}'+\zeta_\phi\phi')\,\mathrm{d}S + \int_M \rho\,\mathbf{u}'\cdot\nabla\psi\,\mathrm{d}V + (\tilde{\mathbf{I}}\boldsymbol{\omega}+\boldsymbol{\Gamma}\dot{\boldsymbol{\omega}}-\mathbf{j}+\mathbf{k})\cdot\boldsymbol{\omega}' = 0, \tag{D3.20}$$

with all loads now functions of t, and eq. (3.21) holding pointwise in t. The steps leading to eqs. (3.22) and (D3.19) combine to give, for $\boldsymbol{\omega}(0) = \mathbf{0}$ and $\boldsymbol{\omega}^\dagger(T) = \mathbf{0}$,

$$\int_0^T \Big\{ \int_{\partial M} \Delta\mathrm{SL}^\dagger\zeta\,\mathrm{d}S - \frac{1}{g}\int_{\partial M}[\mathbf{t}\cdot\mathbf{u}^\dagger + \zeta_\phi(\phi^\dagger+\psi^\dagger)]\,\mathrm{d}S - \frac{1}{g}\,\mathbf{k}\cdot\boldsymbol{\omega}^\dagger \Big\}\,\mathrm{d}t$$
$$= \int_0^T \Big\{ \int_{\partial M} \Delta\mathrm{SL}\,\zeta^\dagger\,\mathrm{d}S - \frac{1}{g}\int_{\partial M}[\mathbf{t}^\dagger\cdot\mathbf{u} + \zeta_\phi^\dagger(\phi+\psi)]\,\mathrm{d}S - \frac{1}{g}\,\mathbf{k}^\dagger\cdot\boldsymbol{\omega} \Big\}\,\mathrm{d}t, \tag{D3.22}$$

and the analogue of eq. (3.23) follows in the same way. Writing δJ as the time integral of eq. (3.22a) gives $\delta J = \int_0^T \int_{\partial M} \Delta \mathrm{SL}^\dagger \delta\zeta \, \mathrm{d}S \, \mathrm{d}t$, where the adjoint problem is driven by $(\zeta^\dagger, \mathbf{t}^\dagger, \zeta_\phi^\dagger, \mathbf{k}^\dagger)(t)$ and solved backwards from $\boldsymbol{\omega}^\dagger(T) = \mathbf{0}$, i.e. forwards in τ with $-\boldsymbol{\Gamma}$.

3.4 Spherically symmetric implementation

With $I_1 = I_2$, the equatorial part of $\boldsymbol{\Gamma}$ is $(I_1/\Omega)\mathsf{E}$, where $\mathsf{E} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ ($\mathsf{E}^T = -\mathsf{E}$, $\mathsf{E}^2 = -1$). The equatorial part of eq. (D3.20), $(I_3 - I_1)\boldsymbol{\omega}_\perp + (I_1/\Omega)\mathsf{E}\dot{\boldsymbol{\omega}}_\perp = \mathbf{j}_\perp - \mathbf{k}_\perp$, becomes, using eq. (P3),

$$\boxed{\dot{\boldsymbol{\omega}}_\perp = \sigma_e \, \mathsf{E} \, (\boldsymbol{\omega}_\perp - \mathbf{q}), \qquad \sigma_e = \frac{\Omega(I_3 - I_1)}{I_1}} \tag{DP3}$$

The static theory (P3) is the limit $\dot{\boldsymbol{\omega}}_\perp \to \mathbf{0}$. In complex notation $w = \omega_1 + i\omega_2$, this is the classical $(i/\sigma_e)\dot{w} + w = q_1 + iq_2$ (Munk & MacDonald 1960), with $e^{+i\nu t}$, $\nu > 0$, prograde.

> **Clarification.** Multiply the equatorial equation by Ω/I_1 to get $\mathsf{E}\dot{\boldsymbol{\omega}}_\perp = \sigma_e(\mathbf{q} - \boldsymbol{\omega}_\perp)$. Then apply $\mathsf{E}^{-1} = -\mathsf{E}$.

Free wobble. By linearity, $\beta\boldsymbol{\phi}_{21} = \mathbf{f}^\zeta(t) + \mathsf{K}\boldsymbol{\omega}_\perp(t)$, where $\mathbf{f}^\zeta$ is driven by the direct load (with the ocean responding) and K by rotation (rotational deformation plus the equilibrium ocean pole tide). Without oceans, $\mathsf{K} = \kappa\mathsf{1}$. Then $\dot{\boldsymbol{\omega}}_\perp = \sigma_e\mathsf{E}[(\mathsf{1}-\mathsf{K})\boldsymbol{\omega}_\perp - \mathbf{f}^\zeta + \mathbf{k}_\perp/(I_3 - I_1)]$. If K is symmetric, as reciprocity suggests, the free frequency is $\sigma_{\mathrm{CW}} = \sigma_e\sqrt{\det(\mathsf{1}-\mathsf{K})}$ and the wobble is undamped. Without oceans, $\sigma_{\mathrm{CW}} = \sigma_e(1-\kappa)$, i.e. $T_{\mathrm{CW}} = 304.5\,\mathrm{d}/(1-0.314) \approx 444\,\mathrm{d}$ (observed: 433 d).

Frequency response. For forcing $\propto e^{i\nu t}$, with $\mathsf{K} = \kappa\mathsf{1}$, $w_{\mathrm{dyn}}/w_{\mathrm{static}} = 1/(1 - \nu/\sigma_{\mathrm{CW}})$:

Period	30 d	182.6 d	365.25 d	∞
prograde	−0.072	−0.70	−4.64	1
retrograde	0.063	0.29	0.45	1

For a load ramp switched on at $t = 0$, the quasi-static trend is correct. The dynamic corrections are a lag of $T_{\mathrm{CW}}/2\pi \approx 70\,\mathrm{d}$, perpendicular to the trend, and a free wobble of the same amplitude as that lag. All these numbers omit the ocean, core and anelastic corrections to σ_{CW} and are illustrative only.

3.5 Time stepping and iterative solution

Trapezoidal step. With $a = \sigma_e\Delta t/2$, $\boldsymbol{\omega}_n = \boldsymbol{\omega}_\perp(t_n)$ and $\mathbf{q}_n = \mathbf{q}(t_n)$,

$$\boldsymbol{\omega}_{n+1} = \mathsf{R}\,\boldsymbol{\omega}_n + \mathsf{S}\,(\mathbf{q}_n + \mathbf{q}_{n+1}), \qquad \mathsf{R} = \frac{(1-a^2)\mathsf{1} + 2a\mathsf{E}}{1+a^2}, \qquad \mathsf{S} = \frac{a^2\mathsf{1} - a\mathsf{E}}{1+a^2}. \tag{DP4}$$

R is an exact rotation. With $\mathsf{K} = \kappa\mathsf{1}$ the full implicit step rotates by $2\arctan[a(1-\kappa)]$, so there is no numerical damping. For monthly steps the period error is 1.5% (450.6 d against 444.0 d). Backward Euler, $\boldsymbol{\omega}_{n+1} = [(1 + 2a\mathsf{E})\boldsymbol{\omega}_n + (4a^2\mathsf{1} - 2a\mathsf{E})\mathbf{q}_{n+1}]/(1 + 4a^2)$, damps the wobble but reproduces the static theory exactly as $\Delta t \to \infty$.

Scheme A: modify the `pyslfp` loop. At step $n+1$, iterate exactly as the static solver does, but replace the update $\boldsymbol{\omega}_\perp \leftarrow \mathbf{q}$ with $\boldsymbol{\omega}_\perp \leftarrow \mathsf{R}\boldsymbol{\omega}_n + \mathsf{S}(\mathbf{q}_n + \mathbf{q}^{(k)})$, where $\mathbf{q}^{(k)}$ is formed from the current $\boldsymbol{\phi}_{21}$ (excluding ψ). After convergence, store $\boldsymbol{\omega}_{n+1}$ and $\mathbf{q}_{n+1}$. The static solver is the case $\mathsf{R} = 0$, $\mathsf{S} = \mathsf{1}$, $\mathbf{q}_n$ term omitted. The rotational sub-iteration contracts with factor at most $\kappa\, a/\sqrt{1+a^2} < \kappa$, so it converges faster than the static loop.

Scheme B: exact superposition. Solve the sea-level equation (i) once for each unit prescribed $\boldsymbol{\omega}_\perp = \mathbf{e}_\alpha$, with ψ imposed but $\boldsymbol{\omega}$ not updated, which gives fields X^α and K; and (ii) at each time for $\zeta(t)$ with $\boldsymbol{\omega} \equiv \mathbf{0}$, which gives $X^\zeta(t)$ and $\mathbf{f}^\zeta(t)$. Then integrate the 2×2 linear ODE exactly or with eq. (DP4); the implicit step is $(1 - \mathsf{SK})\boldsymbol{\omega}_{n+1} = \mathsf{R}\boldsymbol{\omega}_n + \mathsf{S}(\mathbf{q}_n + \mathbf{f}^\zeta_{n+1} - \mathbf{k}_{\perp,n+1}/(I_3 - I_1))$. Finally $X(t) = X^\zeta(t) + \sum_\alpha \omega_\alpha(t) X^\alpha$.

Adjoint. By eq. (D3.19a), run the same solver on time-reversed adjoint loads with $\sigma_e \to -\sigma_e$, i.e. $a \to -a$, $\mathsf{R} \to \mathsf{R}^T$, $\mathsf{S} \to \mathsf{S}^T$. This is the continuous adjoint. The exact discrete adjoint of eq. (DP4), needed for dot-product tests at machine precision, is obtained by transposing the recursion; it agrees to $O(\Delta t^2)$.

3.6 Checks and open issues

- **Checks.** Backward Euler at large Δt reproduces static `pyslfp`; free oscillation at 444 d with constant $|\boldsymbol{\omega}|$ (no load, no ocean); K symmetric and $\mathsf{K} \approx \kappa \mathsf{1}$ without ocean; Schemes A and B agree; space–time dot-product test with eq. (D3.19).
- **Fluid core.** It is largely decoupled from the wobble, so the inertial term involves mantle moments and wobble-appropriate Love numbers (Smith & Dahlen 1981). Pragmatically, tune σ_e to the observed T_{CW}.
- **Anelasticity.** Gives $Q_{\mathrm{CW}} \sim 10^2$. A damping term breaks the time-reversal structure, so the adjoint then carries the opposite-signed damping.
- **Triaxiality** ($I_1 \neq I_2$). Eq. (D3.7) fails; reciprocity in the form (D3.19) is not established here.
- **Observational conventions.** GRACE and altimetry pole-tide corrections use observed polar motion, and should be mirrored consistently in the model (cf. Lickley et al. 2018).

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